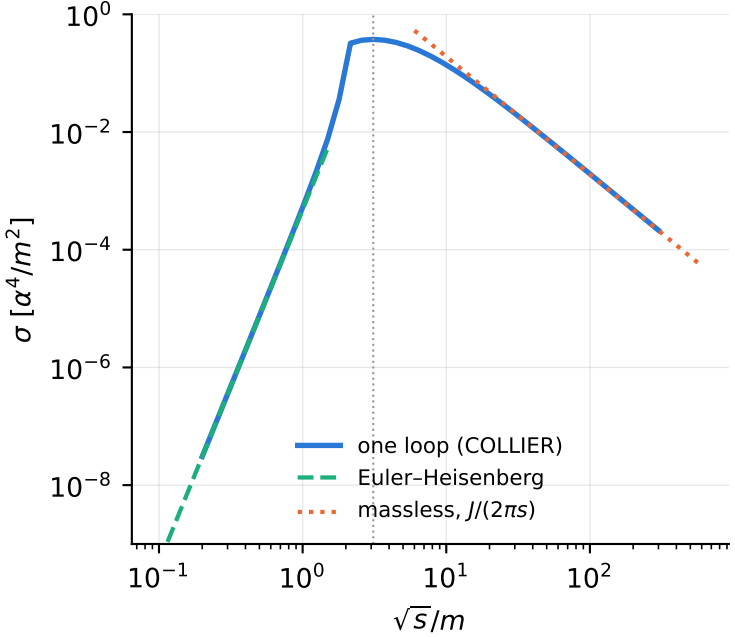
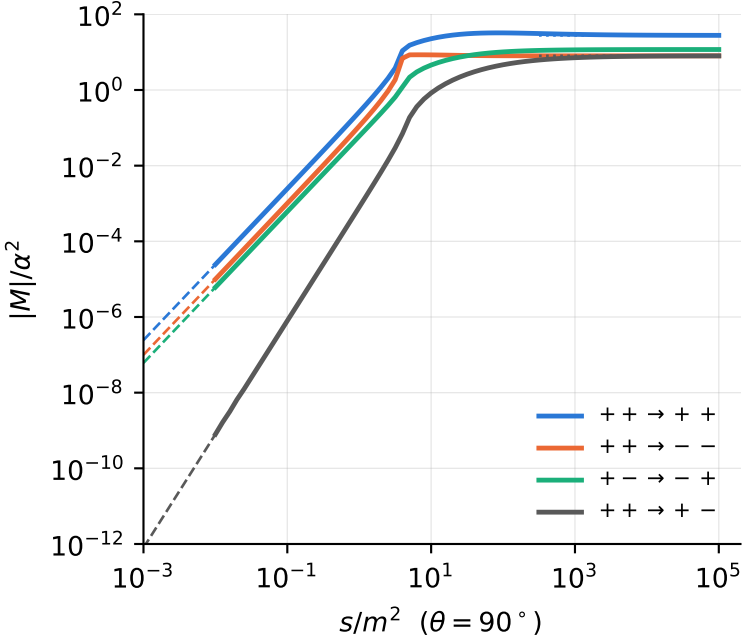
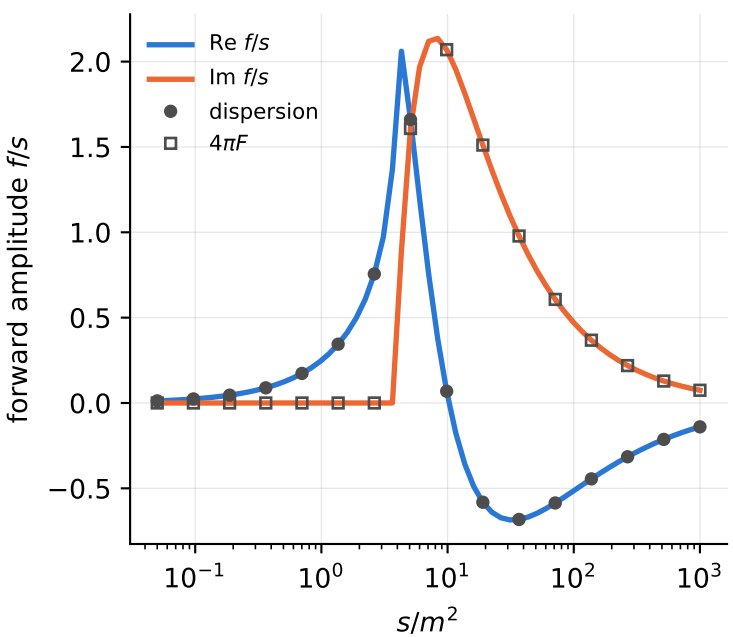

10^{0}
10^{-2}
10^{-4}
10^{-6}
10^{-8}
$\sigma\ [\alpha^4/m^2]$
one loop (COLLIER)
Euler–Heisenberg
massless, $J/(2\pi s)$
10^{-1}
10^{0}
10^{1}
10^{2}
$\sqrt{s}/m$


10^{2}
10^{0}
10^{-2}
10^{-4}
10^{-6}
10^{-8}
10^{-10}
10^{-12}
$|M|/\alpha^2$
$+\,+ \to +\,+$
$+\,+ \to -\,-$
$+\,- \to -\,+$
$+\,+ \to +\,-$
10^{-3}
10^{-1}
10^{1}
10^{3}
10^{5}
s/m^2 $(\theta = 90^\circ)$


Re f/s
Im f/s
dispersion
$4\pi F$
2.0
1.5
1.0
0.5
0.0
−0.5
forward amplitude f/s
10^{-1}
10^{0}
10^{1}
10^{2}
10^{3}
s/m^2